

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-512

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	512	5FA	150A-WW-SLD-ST1-N-8506-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
2	512	5FA	150A-WW-SLD-ST1-N-8507-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
3	512	5FA	150A-WW-SLD-ST1-N-8508-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
4	512	5FA	40A-IW-ST1-C-9535-01	1 OF 2	0	CJ-M15W14-PR-A20-000103	1A	
5	512	5FA	40A-IW-ST1-C-9535-02	2 OF 2	0	CJ-M15W14-PR-A20-000103	1A	

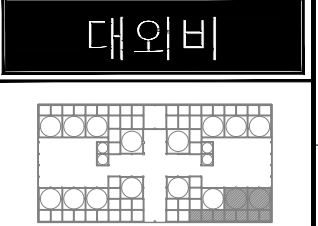
ITEM NO.	W14-T-4107 A
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5) (252.1 (EH 201.1))
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 A
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 5.625 x 8.15 (EH 6.5) (252.1 (EH 201.1))
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 A
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.7 x 5.625 x 8.15 (EH 6.5) (261.3 (EH 208.4))
CAPA.	m3
HRT	min
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 A
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 11.7 x 10.85 x 8.15 (EH 6.5) (1034.6 (EH 825.1))
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304



KEY PLAN (5F)

- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB 으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 거주층의 상계사창은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 검사한 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. VALVE 이용이 접근 가능해야 하며, LIMIT SWITCH가 육안에 보이기 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
REV.	실명	날짜
DRAWN BY	이인학	
CHECKED BY	한진희	
APPROVED BY	박동명	
제작사	SK 에코플랜트	

공사명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
HF 2'nd COAGULATION & SETTLING TANK A (1/2)	
축척	SCALE
NONE	
날짜	DATE
2025. 08. 14	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000103
SHEET NO.	1A

WW-SLD-512

ITEM NO.	W14-AG-4107 A
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(30)	
MAT'L	STS316L/STS316L
POWER	kw
(15)	

ITEM NO.	W14-AG-4108 A
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(25)	
MAT'L	STS316L/STS316L
POWER	kw
(11)	

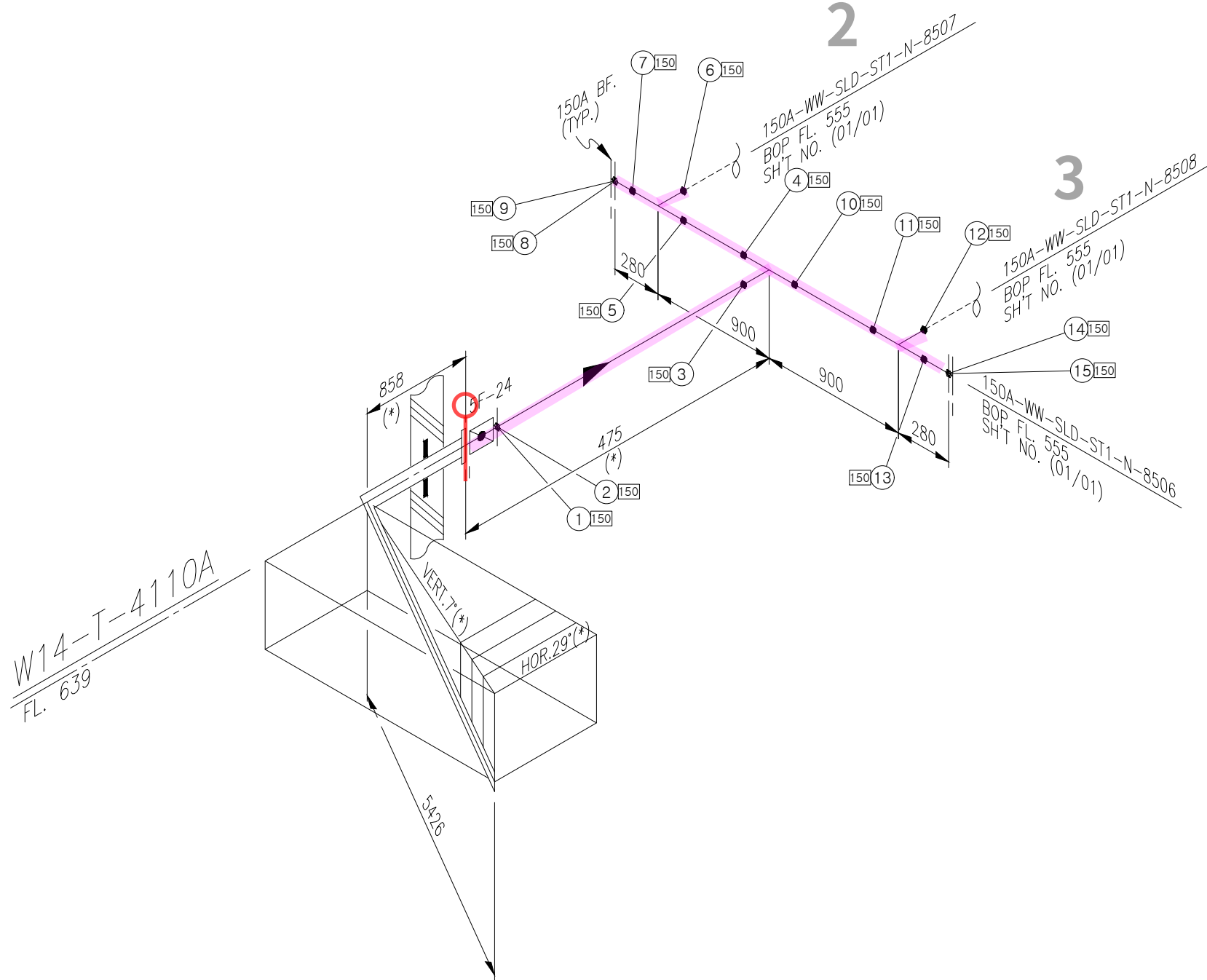
ITEM NO.	W14-AG-4109 A
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm
(25)	
MAT'L	STS316L/STS316L
POWER	kw
(7.5)	

ITEM NO.	W14-RD-4110 A
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d
(0.067)	
MAT'L	STS304/STS304
POWER	kw
(2.2)	

ITEM NO.	W14-P-4110 A1/A2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH
50 x 30 (0.29MPa)	
MAT'L	SSC13/SSC19
POWER	kw
(30)	



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	150A	2.8M	PIPE, STS 304TP, ERW, BE – SCH 10S
T	150A	3EA	TEE EQUAL, STS304, ERW, BE – SCH 10S
VF	150A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /
			TRIM-ST304 / EPDM-LINED, LEVER
F	150A	3EA	FLG SO, KS 10K, RF, STS304
FB	150A	2EA	FLG BLIND, KS 10K, RF, STS304
G	150A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	150A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	150A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



<NOTE>

1. (*) FIELD ADJUST

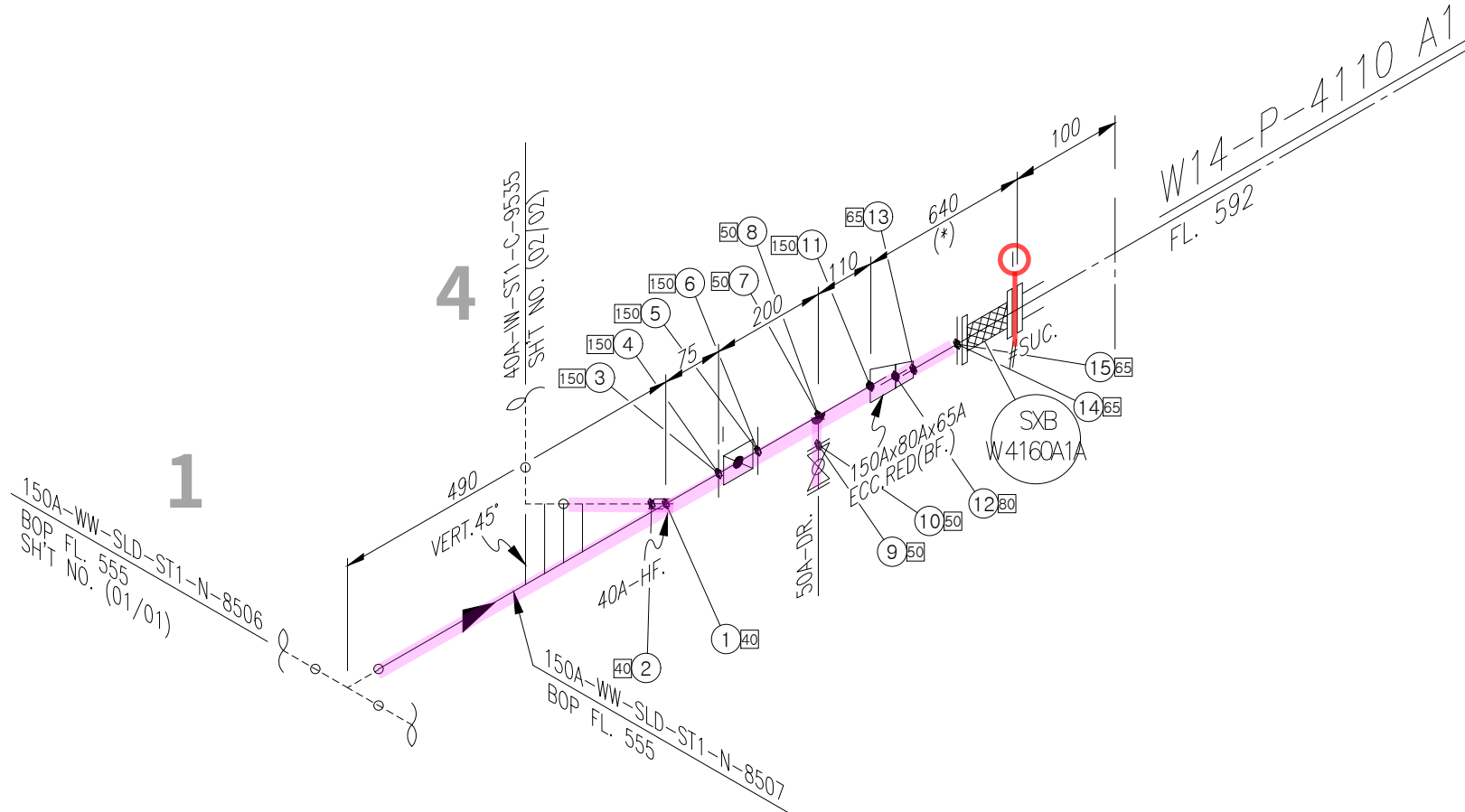
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	0.8	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000103							REV. NO.	DATE	DESCRIPTION	BY

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	150A	0.9M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	65A	0.3M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	50A	0.1M	PIPE, STS 304TP, ERW, BE – SCH 10S
RE	150Ax80A	1EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
RE	80Ax65A	1EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
STU	150Ax50A	1EA	STUB-IN BRANCH
HF	40A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
VB	50A	1EA	BALL VALVE, KS 10K, RF FLC'D, BODY–SSC13 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
VF	150A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY–GCD450 /
			TRIM–STS304 / EPDM–LINED, LEVER
F	150A	2EA	FLG SO, KS 10K, RF, STS304
F	65A	1EA	FLG SO, KS 10K, RF, STS304
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	150A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	150A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
FLXB	65A	1EA	FLEXIBLE JOINT(TPC STEEL TYPE)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR 	PROJECT NAME			
	THICK.	—	DESIGN	0.8	60	UNIT 5FA							M15X Ph4 Project			
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	WW-SLD	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000103							REV. NO.	DATE	DESCRIPTION	BY

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

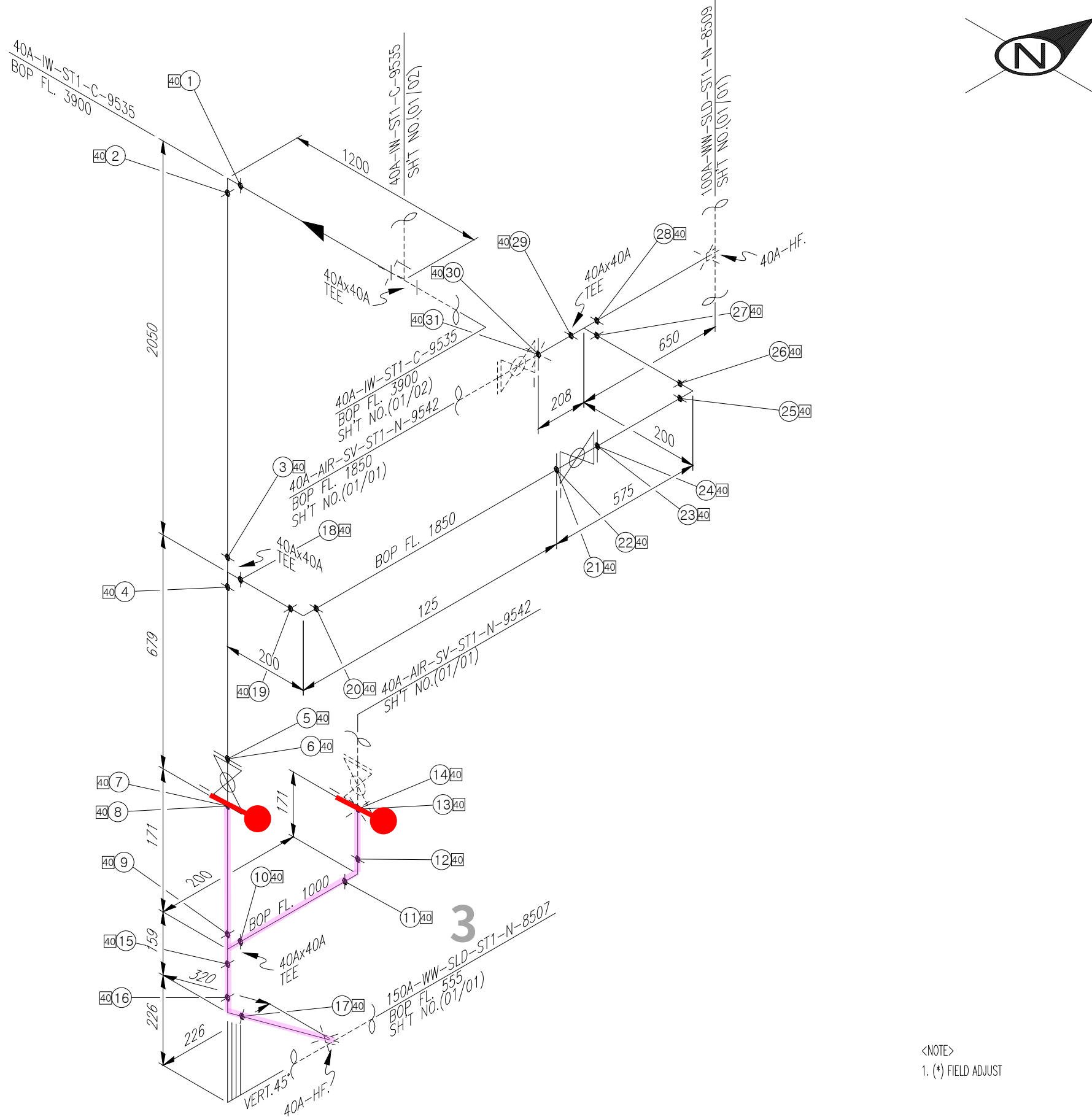


SH'T NO. 01 / 01	REV. 0
---------------------	-----------



INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14
	THICK.	—	DESIGN	0.8	60	UNIT 5FA
			OPERATING	ATM	AMB	REF. PLAN DWG.
						CJ-M15W14-PI-A30-050007-A
						REF. P & ID
						CJ-M15W14-PR-A20-000103

O	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME M15X Ph4 Project	
	THICK.	25	DESIGN	13.5	60	UNIT 5FA									
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	JOB NO.	
						IW							22330D		
						REF. P & ID CJ-M15W14-PR-A20-000103							REV. NO.	DATE	DESCRIPTION
													40A-IW-ST1-C-9535	02 / 02	0



5